Software Requirements Specification (SRS)

Pharma Net: Verified Pharmaceutical Network
Mobile Application and Admin Web App
Version 1.6 | March 19, 2026

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1. Executive Summary

Problem Statement

Pharmacies (Sellers) and distributors/manufacturers (Dealers) in emerging markets face fragmented supply chains, lack of real-time stock visibility, manual ordering delays, expired medicine risks, and unverified suppliers. This leads to stockouts, overstocking, and trust issues.

Proposed Solution

Pharma Net is a **mobile-first Flutter application** (Android + future iOS) + **admin web portal** that creates a verified B2B pharmaceutical network.

- Sellers & Dealers register with OTP + business license upload → admin verification
- Verified users share live inventory, negotiate in real-time chat and view profile
- Public users can search medicines across pharmacies (basic free / advanced, paid once)
- Monetization via product boosting (Sellers pay per product) + one-time advanced search unlock (public users)

Target Audience

- **Sellers** (Retail & Hospital Pharmacies)
- **Dealers** (Pharmaceutical Distributors & Manufacturers)
- **Admins** (Platform moderators for verification & oversight)

2. Functional Requirements & Scope

Core Features

1. **Secure Onboarding & Business Verification** — OTP registration + license upload + admin approval
2. **Real-time Shared Product Catalog & Live Inventory** — Dealers/Sellers publish stock
3. **In-app Chat & Document Sharing** — messaging + payment proof upload

4. **Public Search with Monetized Advanced Filters** — Free basic search + one-time unlock for advanced filters & featured products

User Roles

Role	Can Do	Cannot Do
Seller	Browse catalog, chat, view own	Manage other sellers' inventory
Public	View sellers products, basic and advanced search, view profile, send chat	Create and manage inventory
Admin	Verify/reject users, view all activity, moderate	Send chat message

Out of Scope

- Full online payment processing (only proof upload & manual status)
- iOS release (Flutter prepares for future iOS)
- Multi-language support beyond English & Amharic
- Advanced analytics dashboards & PDF exports
- Pharmacy delivery logistics integration and order management

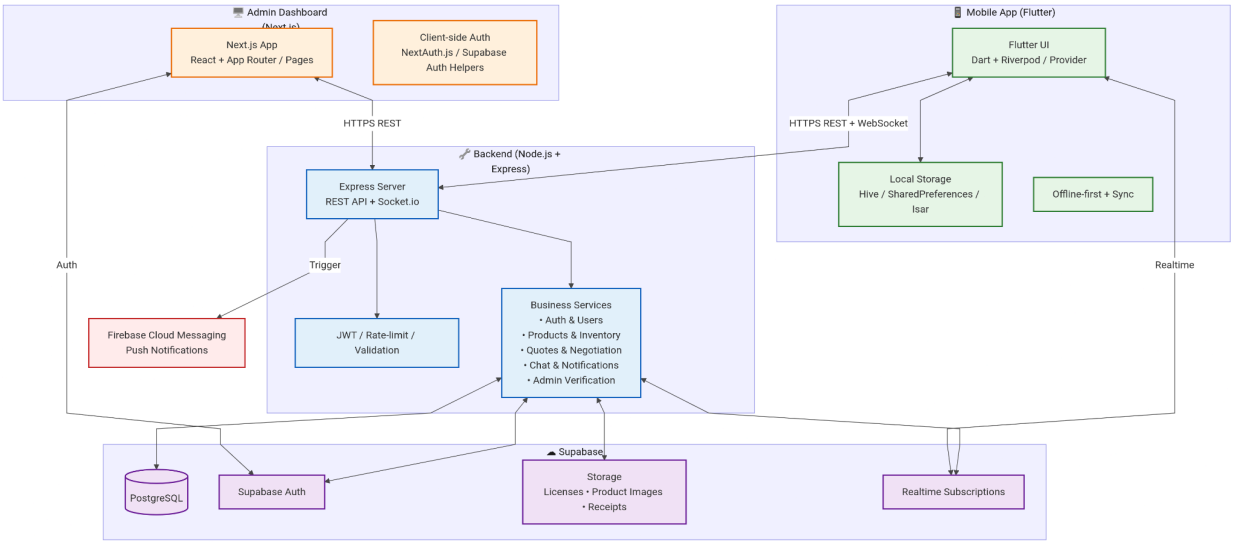
3. System Architecture & Tech Stack

Technology Stack

- **Mobile App** — Flutter (Dart) + Riverpod / Provider
- **Admin Portal** — Next.js (React + App Router)
- **Backend** — Node.js + Express.js + Socket.io
- **Database** — PostgreSQL via Supabase
- **Authentication** — Supabase Auth (OTP + JWT)
- **Storage** — Supabase Storage (licenses, product images, receipts)
- **Real-time** — Supabase Realtime + Socket.io (chat)
- **Push Notifications** — Firebase Cloud Messaging (FCM)
- **Local Caching (Mobile)** — Hive / Isar / SharedPreferences

High-Level Architecture Diagram

Note: If not visible? View HL Architecture diagram in [Google Drive] in full resolution



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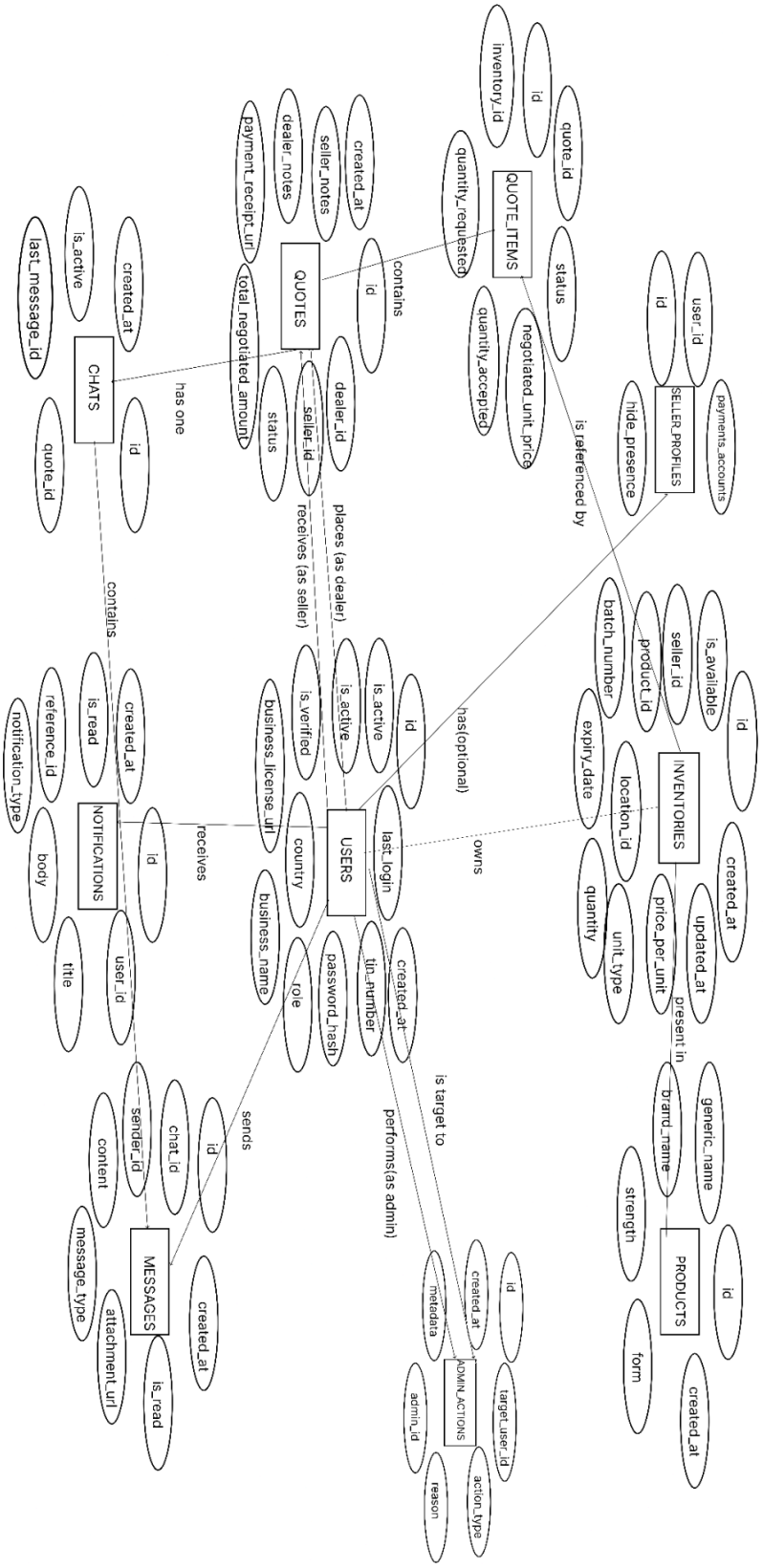
4. Database Design

ER Diagram and Schema Table

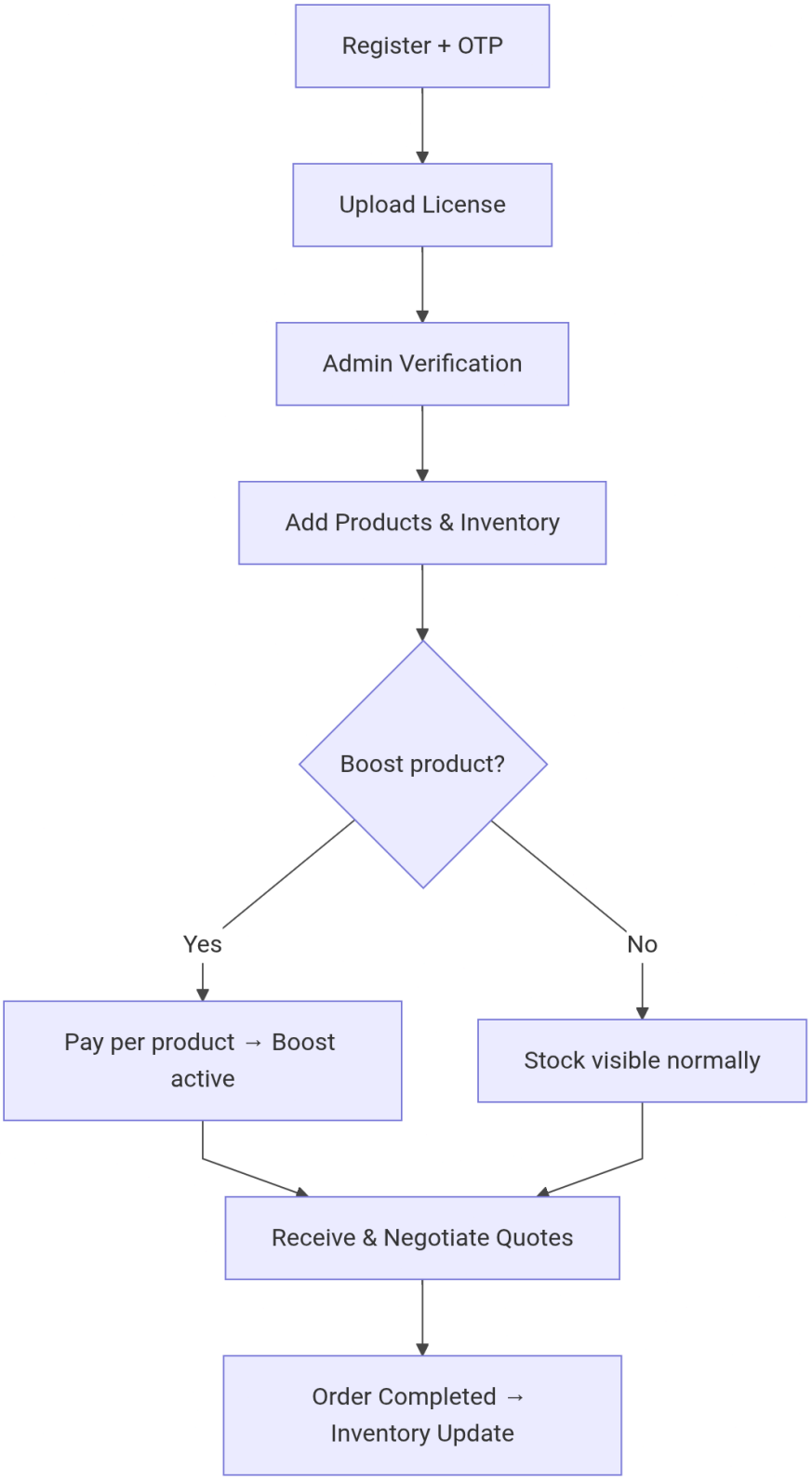
Note: If not visible? View ER diagram in [Google Drive] in full resolution our in our [figma file]



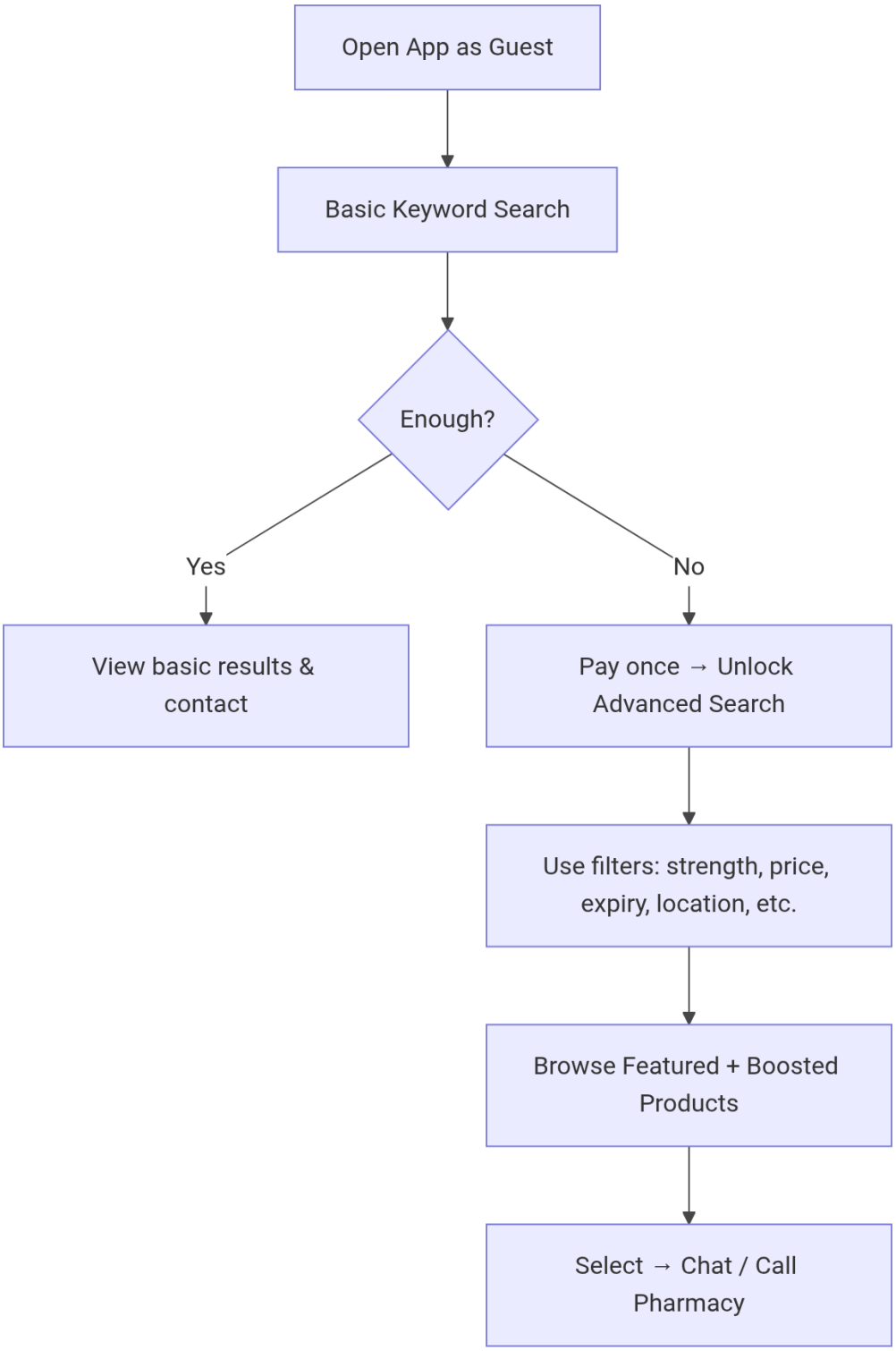
Note: If not visible view it in [Google Drive]() in full resolution or in our [figma file]()

5. User Flow & Wireframes

User Flow Diagram (Seller) / Business Flow



User Flow Diagram (Public) / Business Flow



Key Wireframes and UI/UX

Full detailed wireframes & UI/UX: View in Figma

6. Implementation Plan

Milestones (4 Phases – 8 Weeks Total)

Phase 1: Foundation (Weeks 1–2)

- Node.js + Express project setup
- Supabase schema & authentication
- Mobile app skeleton + OTP login + license upload

Phase 2: Core Features (Weeks 3–4)

- Backend APIs (Products, Inventory, Chat)
- Mobile catalog + real-time stock

Phase 3: Communication & Polish (Weeks 5–6)

- In-app chat + notifications (FCM)
- Offline support + sync
- Admin verification panel

Phase 4: Testing & Launch (Weeks 7–8)

- Full end-to-end implementation
- Performance optimization
- Beta release + final APK generation

Definition of Done

- Fully functional Android APK available for download
- End-to-end flow works: Register → Verify → Browse → Chat
- All core features tested on real devices
- Backend deployed and stable
- Business license verification process operational